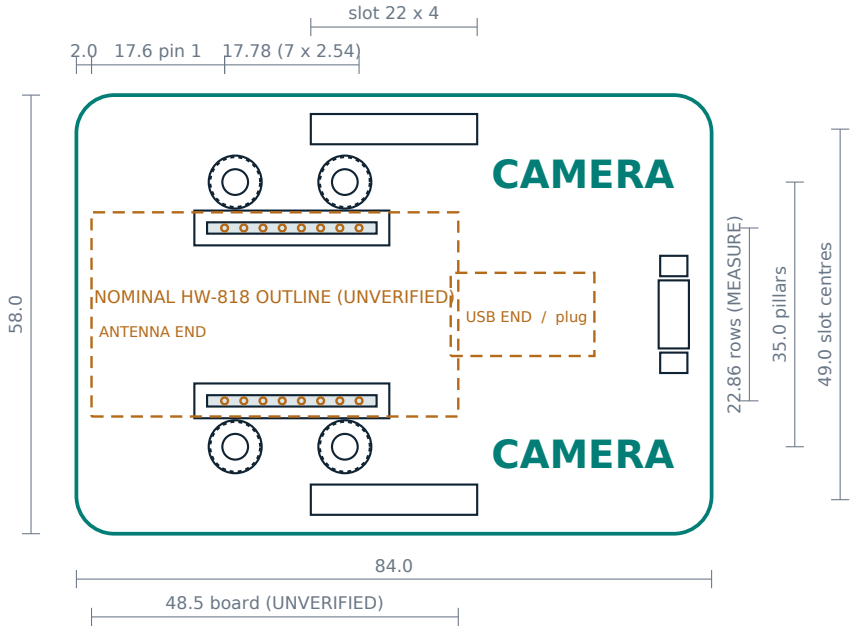
MicroScan V5 | Sheet 1 of 3 | Carrier plan view, scale 1:1

Third-angle. Dimensions in mm. General printed tolerance +-0.3; holes modelled +0.4 on diameter for FDM.

BOARD DIMENSIONS ARE UNVERIFIED (seller datasheet + listing photo). NOT FABRICATED. NOT FIT-TESTED.



- Datum: plate centre (0,0). Board antenna-end edge at X = -40.0; USB-end edge at X = 8.5.
- M3 pillars (X, Y): (-21.0, -17.5), (-21.0, 17.5), (-6.5, -17.5), (-6.5, 17.5); dia 7.0, hole dia 3.4.
- Skin-side pan-head pockets (dashed): dia 6.6 x 2.6 deep. Head must sit below the face.
- Pin troughs: 1.5 wide, blind, floor 1.8 solid toward skin. Never drill through.
- Cable saddle at X = 37.0; tie windows 3.6 x 2.7 at Y = +-6.4; tie lies in a 1.2 deep skin-side channel.
- Engraved "CAMERA" x2, 0.8 deep. Fill with contrasting paint. Do not cover the lens, the label or the capture LED.
- One 20 mm hook-and-loop strap through the two long-edge slots. Cable stays outside clothing and fully visible.

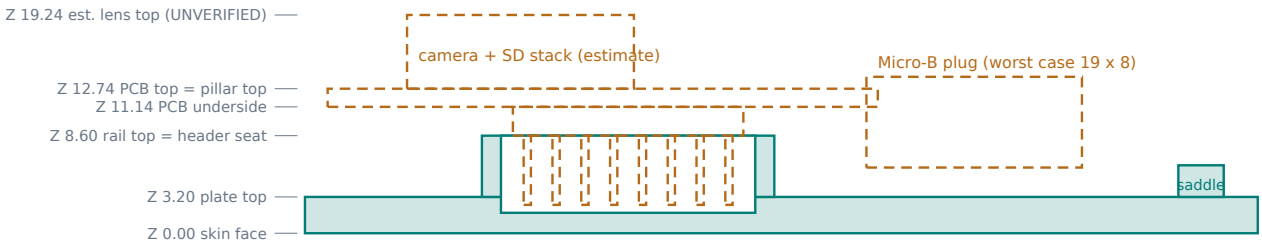
Scale check: this ruler must measure 100.0 mm. If not, reprint at 100 %.



MicroScan V5 | Sheet 2 of 3 | Section through a pin row, scale 1.5:1

Section plane Y = +11.43 (pin row). Vertical datum Z = 0 is the skin-side face.

BOARD DIMENSIONS ARE UNVERIFIED (seller datasheet + listing photo). NOT FABRICATED. NOT FIT-TESTED.



HEIGHT STACK (nominal): plate 3.2 + rail 5.4 + header plastic 2.54 + PCB 1.6 (ASSUMED) = 12.74 to PCB top.

Pin tips end 0.70 above the blind trough floor; 1.8 mm of solid plate stays between pins and skin.

Open air gap under the ESP-32S can: 4.74 mm. Under the Micro-B overmold: 2.59 mm. Nothing encloses the board.

FASTENERS x4: M3 pan head, head in skin-side pocket, shank up through pillar and retainer slot, M3 nut on top.

Grip (head seat to retainer top) = 13.14 mm. Nut 2.4. M3x16: 0.46 mm thread past nut. M3x20: 4.46. M3x15: TOO SHORT.

Nylon (PA66) screws and nuts preferred. Steel only if the nut cannot reach a solder joint; excess thread points away from skin.

IF THE REAL PCB IS THINNER THAN 1.6: add an M3 nylon washer or 0.5 mm shim under the retainer tip; never over-tighten.

IF HEADERS ARE NOT FITTED, OR POINT TO THE CAMERA FACE: STOP. This carrier does not apply; re-measure and re-parametrise.

PRINT: PETG, 0.2 mm layers, 0.4 nozzle, 4 perimeters, 5 top/bottom layers, >= 40 % infill, skin face on the bed, no supports

(pocket ceilings are 6.6 mm bridges). PLA softens near 55-60 C and is not recommended next to a 5 V regulator.

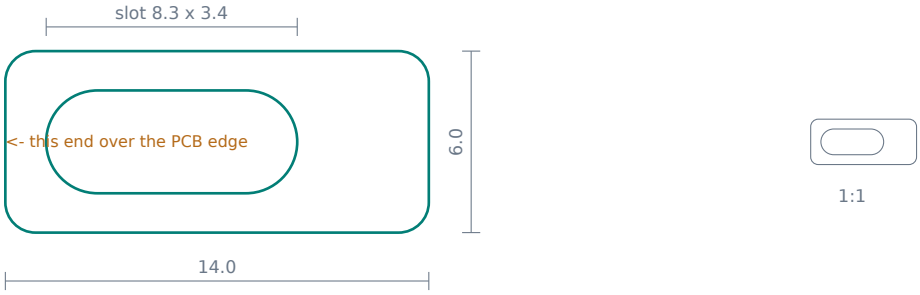
Scale check: this ruler must measure 100.0 mm. If not, reprint at 100 %.



MicroScan V5 | Sheet 3 of 3 | Retainer 4:1 and 1:1, PCB gauge 1:1

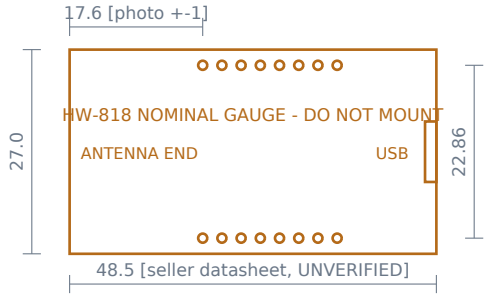
Cut out or print the gauge outline; lay the real board on it BEFORE printing the carrier.

BOARD DIMENSIONS ARE UNVERIFIED (seller datasheet + listing photo). NOT FABRICATED. NOT FIT-TESTED.



thickness 3.0; fillet R1; print flat; qty 4

Edge overlap adjustable -0.95 ... 3.95 mm about the bolt. SET 1.0 mm: more than ~1.15 mm reaches the header solder fillets.



MEASURE ON THE REAL BOARD AND WRITE HERE BEFORE PRINTING (edit the same names in cad/build_hardware_v5.py):

PCB_L _____ PCB_W _____ PCB_T _____ HDR_ROW_SPACING _____ HDR_PIN1_FROM_ANT_END _____

HDR_PIN_BELOW_BODY _____ headers fitted, pins away from camera face? YES / NO

USB receptacle on the module face at the end opposite the antenna? YES / NO plug overmold L x W x H _____

tallest part under the PCB _____ (must be < 7.9) PCB silk says HW-818? YES / NO

Accept the carrier only if every pin drops into a trough without force and the board sits flat on both rails.

Scale check: this ruler must measure 100.0 mm. If not, reprint at 100 %.

